

CC2 - Test-out Assessment Blueprint

Primary Standard	Source	DOK	Weight (points)	Item #
7.RP.2a 2.Recognize and represent proportional relationships between quantities. a.Decide whether two quantities are in a proportional relationship, e.g., by testing for equivalent ratios in a table or graphing on a coordinate plane and observing whether the graph is a straight line through the origin.	CC2 Chapter 4 Individual assessment Adapted Item #4a	2	2	
	CC2 Chapter 5 Individual assessment Adapted Item #6c	2	2	
	CC2 Chapter 8 Review/Preview Adapted 8-78	2	2	
	North Carolina Instructional Support Tools Grade 7 document	2	2	
8 points				
7.RP.2b 2.Recognize and represent proportional relationships between quantities. b.Identify the constant of proportionality (unit rate) in tables, graphs, equations, diagrams, and verbal descriptions of proportional relationships.	CC2 Chapter 7 Individual assessment Item #6 b and c	2	3	
	CC2 Chapter 9 Review/Preview 9-50	2	4	
	CC2 Chapter 9 Review/Preview 9-92 a and b	2	4	
11 points				
7.SP.5 5. Understand that the probability of a chance event is a number between 0 and 1 that expresses the likelihood of the event occurring. Larger numbers indicate greater likelihood. A probability near 0 indicates an unlikely event, a probability around 1/2 indicates an event that is neither unlikely nor likely, and a probability near 1 indicates a likely event.	CC2 Chapter 3 Individual assessment Adapted Item #5	2	2	
	CC2 Chapter 5 Review/Preview 5-29 b and c	2	2	
4 points				
7.NS.1d 1. Apply and extend previous understandings of addition and subtraction to add and subtract rational numbers; represent addition and subtraction on a	North Carolina Instructional Support Tools Grade 7 document	2	6 (2 each)	

horizontal or vertical number line diagram. d. Apply properties of operations as strategies to add and subtract rational numbers. 7 points	Smarter Balanced 7.NS.1d sample item	2	1	
7.NS.2c 2. Apply and extend previous understandings of multiplication and division and of fractions to multiply and divide rational numbers. c. Apply properties of operations as strategies to multiply and divide rational numbers. 8 points	North Carolina Instructional Support Tools Grade 7 document	2	8 (2 each)	
7.EE.4a 4. Use variables to represent quantities in a real-world or mathematical problem, and construct simple equations and inequalities to solve problems by reasoning about the quantities. a. Solve word problems leading to equations of the form $px + q = r$ and $p(x + q) = r$, where p, q, and r are specific rational numbers. Solve equations of these forms fluently. Compare an algebraic solution to an arithmetic solution, identifying the sequence of the operations used in each approach. <i>For example, the perimeter of a rectangle is 54 cm. Its length is 6 cm. What is its width?</i> 6 points	CC2 Chapter 6 Individual Assessment Adapted item #8	2	2	
	CC2 Chapter 7 Individual Assessment Adapted item #2	2	2	
	CC2 Chapter 9 Individual Assessment Item #8	2	2	
7.G.4 Know the formulas for the area and circumference of a circle and use them to solve problems; give an informal derivation of the relationship between the circumference and area of a circle. 4 points	CC2 Chapter 9 Individual assessment Item #4	2	2	
	CC2 Adapted from CPM Test Generator Item Bank	2	2	
7.G.6 6. Solve real-world and mathematical problems involving	CC2 Adapted from CPM Test Generator Item	2	2	

area, volume and surface area of two- and three-dimensional objects composed of triangles, quadrilaterals, polygons, cubes, and right prisms. 6 points	Bank			
	CC2 Chapter 9 Individual assessment Item # 4	2	2	
	North Carolina Instructional Support Tools Grade 7 document	2	2	

1 pt. for the correct answer

1 pt. for showing work that could lead to the correct answer

1 pt. for an explanation that demonstrates understanding of the problem

7.RP.2a

7.RP.2a

Chapter 4 Individual Assessment

4. Last week, Marcy bought 2 pounds of pistachio nuts and paid \$7.96. This week she bought only $\frac{1}{2}$ of a pound and paid \$1.99.

YES or NO: Is the relationship between the pounds of pistachio nuts and the cost a proportional relationship? Show your work.

1 pt. for the correct answer

1 pt. for showing work that could lead to the correct answer

Chapter 5 Individual Assessment

7.rp.2

6. YES or NO: Do the values in the table represent a proportional relationship? Show your work.

x	3	4	5	6	7
y	22	28	34	40	46

1 pt. for the correct answer

1 pt. for showing work that could lead to the correct answer

Review/Preview 8-78

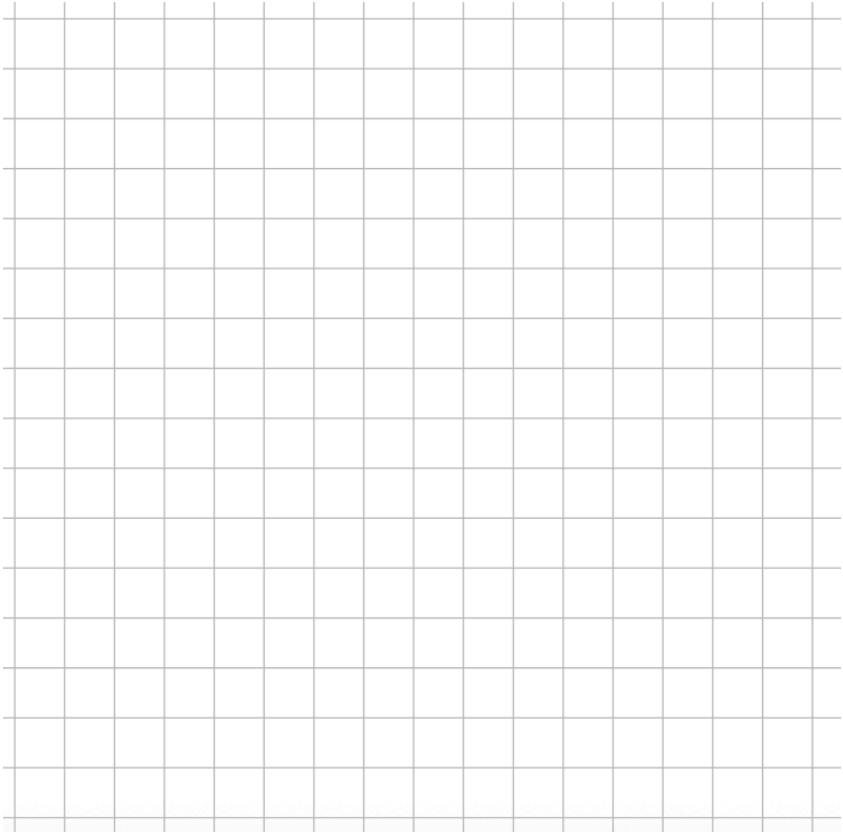
A candy store’s specialty is taffy. Customers can fill a bag with taffy and the price is based on how much the candy weighs. The store charges \$2.00 for 10 ounces of taffy.

A. Complete the table.

Amount of taffy (ounces)	2	5	10	12	15	20
Price (\$)			\$2.00			\$4.00

1 pt. for the correct answers

B. Graph the values from the table you completed above. Let x represent the number of ounces and y represent the price in dollars. 1 pt. for the correct answer



Circle the table that does NOT represent a proportional relationship. Explain why it is not proportional.

1 pt. for the correct answer

1 pt. for a viable explanation

x	y
0	0
1	5
3	15
4	20

x	0	1	3	5
y	0	1	9	25

7.RP.2b

Chapter 7 Individual Assessment

(7.RP.A.2,

6. Greg has been buying apples for 69¢ each from the corner grocer for years. Recently, the grocer started charging 5¢ for each plastic bag the customer takes to put his apples in. Greg *always* takes just one plastic bag, even when it seems like he has a lot of apples.

- a. If x represents the number of apples a customer is buying, and y is the total cost, write an equation representing this situation.

1 pt. for the correct answer

- b. Could Greg ever be charged \$4.32 for his apples? Why or why not? Explain.

1 pt. for the correct answer

1 pt. for a viable explanation

Chapter 9

9-50. Review/Preview

Two employees of Frontier Fence Company can install 100 feet of fence in two days.

- A. What is the unit rate for one employee?

1 pt. for the correct answer

- B. Write a proportion to show the relationship between this rate and 150 feet of fence.

1 pt. for the correct answer

- C. At this same rate, how many employees are needed to install 150 feet of fence in one day? Show your work.

1 pt. for the correct answer

1 pt. for showing work that could lead to the correct answer

Chapter 9

Review/Preview 9-92

Solve.

- a. Tina drove 108 miles and used 4.5 gallons of gasoline. Compute the unit rate. Show your work.

1 pt. for the correct answer

1 pt. for showing work that could lead to the correct answer

- b. If 17 oranges cost 3.23, compute the unit rate. Show your work.

1 pt. for the correct answer

1 pt. for showing work that could lead to the correct answer

7.SP.5

Chapter 3 individual assessment

7.SP.5

5. You have five blue candies, six green candies, nine yellow candies, and three red candies in a bag. You shake the bag, reach in, and pull out a candy at random. What is the probability that the candy will be green? Show your work.

1 pt. for the correct answer

1 pt. for showing work that could lead to the correct answer

Chapter 5

Review/Preview 5-29 b and c

Find the probability of each event.

- a. Rolling a number less than five on a standard number cube.

1 pt. for the correct answer

- b. Drawing a blue marble from a bag of 18 marbles, three of which are blue.

1 pt. for the correct answer

7.NS.1d

North Carolina Instructional Support Tools - Sample Item

Evaluate the expressions. Show your work.

$$5\frac{1}{2} + (-3.25)$$

$$-283 + (-35)$$

$$-4\frac{1}{4} - (6\frac{1}{3})$$

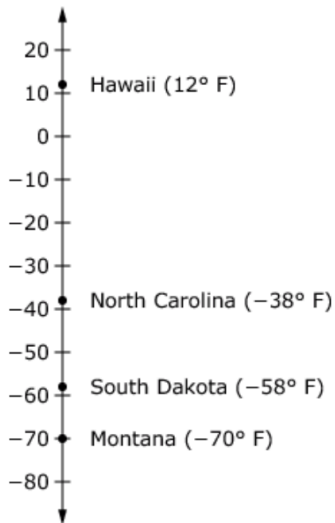
1 pt. for the correct answer

1 pt. for showing work that could lead to the correct answer

(6 points total)

Smarter Balanced Sample Item

The number line shows record low temperatures for four states.



What is the difference, in degrees, between the record low temperatures in Hawaii and South Dakota?

1 pt. for the correct answer

7.NS.2c

North Carolina Instructional Support Tools - Sample Item

Evaluate the expressions. Show your work.

$$-\frac{3}{4} \div \frac{1}{2}$$

$$-5.25 \div (-5)$$

$$\frac{5}{6}(-66)$$

$$-2(10)(-3)$$

1 pt. for the correct answer

1 pt. for showing work that could lead to the correct answer

7.EE.4a

CPM Chapter 6 Individual Test #8 adapted

Solve. Show your work.

Ted weighs 30 pounds. Royce weighs 3 pounds more than twice Ted's weight. How much does Royce weigh?

1 pt. for the correct answer

1 pt. for showing work that could lead to the correct answer

Chapter 7 Individual Assessment

(7.EE.B.4)

2. Solve. Show your work.

Tasha has eight more fish than Oscar, who has twice as many fish as Cecilia. If altogether they have 148 fish, how many fish do each of them have?

1 pt. for the correct answer

1 pt. for showing work that could lead to the correct answer

Chapter 9 individual assessment

7. EE.4

8. Solve. Show your work.

The length of a rectangular field is 7 meters less than four times its width. The perimeter is 136 meters. Calculate the length and the width of the field. ~~Use the 5-D process if you think it will help you.~~

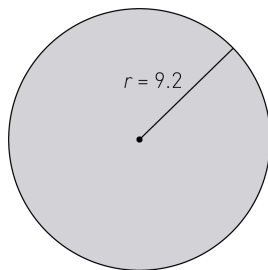
1 pt. for the correct answer

1 pt. for showing work that could lead to the correct answer

7.G.4

Chapter 9 Individual Assessment

Calculate the circumference of the circle. Show your work.



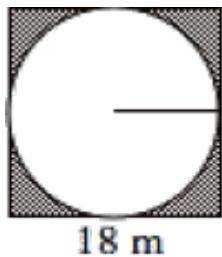
Circumference = _____

1 pt. for the correct answer

1 pt. for showing work that could lead to the correct answer

CPM assessment bank

What is the area of the shaded region within the square?



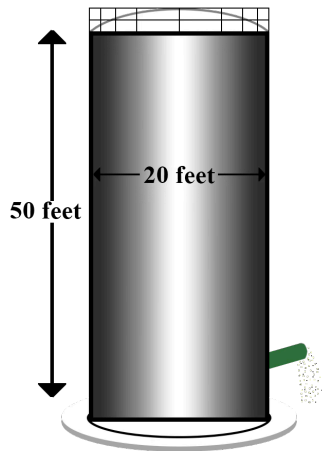
1 pt. for the correct answer

1 pt. for showing work that could lead to the correct answer

7.G.6

(7.G.B.6)

Below is a picture of a cylindrical grain silo.



b. What is the volume of the cylinder given the height is 50 feet? Round your answer to the nearest whole number.

1 pt. for the correct answer

1 pt. for showing work that could lead to the correct answer

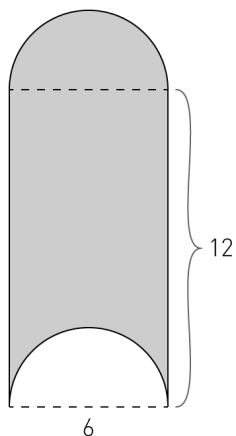
Chapter 9 Individual Assessment

Steve's recommendation is to leave it as is because none of the other geometry items indicate anything about being drawn to scale.

Calculate the area of the shaded portion of the shape below.

1 pt. for the correct answer

1 pt. for showing work that could lead to the correct answer



From North Carolina:

Huong covered the box to the right with sticky-backed decorating paper. The paper costs 3¢ per square inch. How much money will Huong need to spend on paper?

